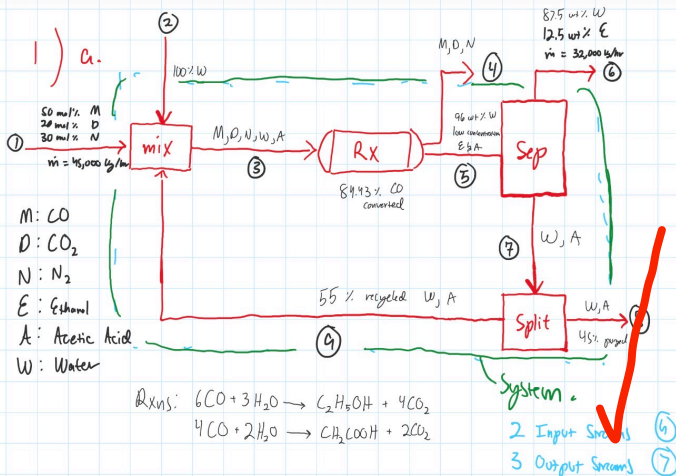




DOF:

4 a	11	5 a	2
4 b	2	5 b	4
	<u>13</u>	5 c	1
		5 d	6
		<u>13</u>	

DOF for overall System.

2 Input Streams
3 Output Streams

$13 - 13 = 0$ — Well Posed!

b. Molar Flow rates in all 9 Streams

Stream 1: MW: $0.5(28) + 0.2(44) + 0.3(28) = 31.2 \text{ kg/kmol}$

$\frac{45000 \text{ kg/hr}}{31.2 \text{ kmol/hr}} = 1442.3 \text{ kmol/hr}$

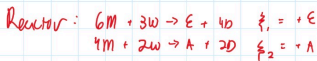
$\dot{m}_1 = 0.5(1442.3) = 721.2 \text{ kmol/hr}$

$\dot{m}_2 = 0.2(1442.3) = 288.5 \text{ kmol/hr}$

$\dot{m}_3 = 0.3(1442.3) = 432.7 \text{ kmol/hr}$

Stream 6: $\dot{E}_6 = \frac{32,000(0.125)}{44} = 86.96 \text{ kmol/hr}$

$\dot{W}_6 = \frac{32,000(0.875)}{18} = 1555.6 \text{ kmol/hr}$



$$\xi_1 = \dot{E}_5 = \dot{E}_6 = 86.96 \text{ kmol/hr}$$

$$M_1 = M_3 = 721.2 \text{ kmol/hr} \quad \text{Conversion } Z$$

$$0.54M_3 = \frac{721.2 - M_4}{721.2}$$

$$M_4 = 112.291 \text{ kmol/hr}$$

$$M_4 = M_3 - 6\xi_1 - 4\xi_2$$

$$112.29 = 721.2 - 6(86.96) - 4\xi_2$$

$$85.15 = 4\xi_2$$

$$\xi_2 = 21.3$$

Stream 5: $\dot{E}_5 = \xi_1 = 86.96 \text{ kmol/hr}$

Splitter:

$$\begin{cases} \dot{A}_5 = \dot{A}_3 + \xi_2 \\ \dot{A}_5 = \dot{A}_7 \\ 0.45\dot{A}_7 = \dot{A}_8 \quad \} \quad 0.55\dot{A}_7 = \dot{A}_9 \\ \dot{A}_3 = \dot{A}_9 \end{cases}$$

$$\dot{A}_5 = 6.55\dot{A}_5 + 21.3$$

$$\dot{A}_5 = 48.42 \text{ kmol/hr}$$

$$\dot{W}_5 \rightarrow 0.96 = \frac{18(\dot{W}_5)}{18(\dot{W}_5) + 46(86.96) + 6(48.42)}$$

$$\dot{W}_5 = 9214.6 \text{ kmol/hr}$$

Stream 7: $\dot{A}_7 = \dot{A}_5 = 48.42 \text{ kmol/hr}$

Water: $\dot{W}_7 = \dot{W}_5 - \dot{W}_6$
 $= 9214.6 - 1555.6$

$$\dot{W}_7 = 7659.0 \text{ kmol/hr}$$

Stream 8: $\dot{W}_8 = 0.45 (\dot{W}_7) = 3446.6 \text{ kcal/hr}$
 $\dot{A}_8 = 0.45 (\dot{A}_7) = 21.79 \text{ kcal/hr}$

Stream 9: $\dot{W}_9 = 0.55 (\dot{W}_7) = 4212.5 \text{ kcal/hr}$
 $\dot{A}_9 = 0.55 (\dot{A}_7) = 26.63 \text{ kcal/hr}$

Stream 3: $\dot{M}_3 = \dot{M}_1 = 7212 \text{ kcal/hr}$
 $\dot{O}_3 = \dot{O}_1 = 288.5 \text{ kcal/hr}$
 $\dot{N}_3 = \dot{N}_1 = 432.7 \text{ kcal/hr}$
 $\dot{A}_3 = \dot{A}_9 = 26.63 \text{ kcal/hr}$

$\dot{W}_5 = \dot{W}_3 - 3\dot{F}_1 + 2\dot{F}_2$
 $\dot{W}_3 = \dot{W}_5 + 3\dot{F}_1 - 2\dot{F}_2$
 $= 4214.6 + 3(6646) - 2(2179)$
 $\dot{W}_3 = 9519.1 \text{ kcal/hr}$

Stream 2: $\dot{W}_3 = \dot{W}_2 + \dot{W}_9$

$\dot{W}_2 = \dot{W}_3 - \dot{W}_9$
 $= 9519.1 - 4212$
 $\dot{W}_2 = 5306.6 \text{ kcal/hr}$

Stream 4: $\dot{m}_4 = \dot{m}_3 - 6\dot{F}_1 + 4\dot{F}_2$
 $\dot{m}_4 = 112.3 \text{ kcal/hr}$
 $\dot{D}_4 = \dot{D}_3 + 4\dot{F}_1 + 2\dot{F}_2$
 $\dot{D}_4 = 679.9 \text{ kcal/hr}$
 $\dot{N}_4 = \dot{N}_3 = 432.7 \text{ kcal/hr}$

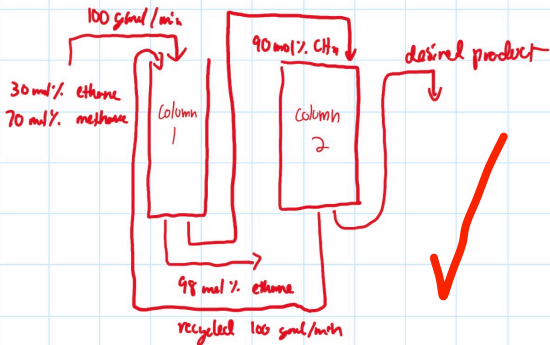
C. Single Pass conversion = $\frac{9519.1 - 4214}{9519.1} \times 100 = 3.2\%$

Overall Conversion = $\frac{5306.6 - 1556 - 546.6}{5306.6} \times 100 = 5.74\%$

To improve water efficiency of process

- ① Reduce purge rate but make sure there is no too much Acetic Acid build-up
- ② Improve separation less water goes to purge and remove more acetic acid before split

2) 5.32



$$F_1 = 100 \text{ g/mol/min} \begin{cases} \dot{m}_1 = 70 \text{ g/mol/min} \\ \dot{E}_1 = 30 \text{ g/mol/min} \end{cases}$$

Overhead: $\dot{V}_1 = x_{m_1} = 0.90 \quad x_{E_1} = 0.10$

Bottoms: $\dot{L}_1 = y_{E_1} = 0.98 \quad y_{m_1} = 0.02$

$\dot{V}_1 \rightarrow$ overhead: $x_{m_2} = 0.97 \quad x_{E_2} = 0.03$ (Product)

Bottoms: $F_2 = 100 \text{ g/mol/min} \quad y_{m_2}, y_{E_2}?$ (recycled)
Col 1

$$V_1 = P + 100$$

$$0.9 V_1 = 0.97P + 100 X_{mg}$$

$$0.9P + 90 = 0.97P + 100 X_{mg}$$

$$100 X_{mg} = 90 - 0.07P$$

$$F = F_1 + F_2 = 200 \text{ gml/min}$$

$$200 = V_1 + V_2$$

$$L_1 = 200 - V_1$$

$$= 200 - (P + 100)$$

$$L_1 = 100 - P$$

$$\text{Methane in} = 20 + 100 X_{mg}$$

$$\text{out} = 0.90 V_1 + 0.62 L_2$$

$$88P + 92$$

$$20 + 100 X_{mg} = 0.88P + 92$$

$$20 + (90 - 0.07P) = 0.88P + 92$$

$$P = 71.6 \text{ gml/min}$$

$$100 X_{mg} = 90 - 0.07(71.6)$$

$$X_{mg} = 0.15$$

$$\text{Product Flow rate} = 71.6 \text{ gml/min}$$

$$\text{Reynolds Compositions} = X_{mg} = 0.85$$

$$X_{EA} = 0.15$$

$$\text{Fractional Recovery: methane} = \frac{0.97(71.6)}{0.7(100)} \times 100 = 99.2\%$$

$$\text{ethane} = \frac{0.98(28.4)}{0.30(100)} \times 100 = 92.8\%$$

3930 ✓

3) 5.34

Feed: $z_E = 0.30$ $z_W = 0.7$

Pressure: 1 atm

Flash $V = L$ (50% vapor, 50% liq.)

$$z_E = \frac{x_E + y_E}{2} = 0.3$$

$$x_E + y_E = 0.60$$

From Table: @ $T = 85.3^\circ$ $x_E + y_E = 0.597$

@ $T = 89.1^\circ$ $x_E + y_E = 0.675$

Interpolate $f \frac{0.60 - 0.597}{0.675 - 0.597} \approx 0.078$

$$T \approx 85.3 - 0.078(85.3 - 89.1)$$

$$T = 85.2^\circ$$

$$x_E = 0.1235 + 0.678(0.164 - 0.1235)$$

$$x_E = 0.1268$$

$$x_W = 0.523$$

$$y_E = 0.4704 + 0.078(0.589 - 0.4704)$$

$$y_E = 0.4732$$

$$y_W = 0.527$$

Fractional recovery:
of ethanol in water

$$0.5(0.4732) = 0.2366$$

$$\frac{0.2366}{0.30} \times 100 = 78.9\%$$

Fractional recovery:
of water in liquid

$$0.5(0.4732) = 0.2366$$

$$\frac{0.2366}{0.70} \times 100 = 33.8\%$$

Separation factor:

$$\frac{(y_E/x_E)}{(y_W/x_W)} = \frac{(0.4732/0.1268)}{(0.527/0.573)}$$

$$S = 6.2$$